

SMART IoT-BASED PLANT IRRIGATION SYSTEM

University Project Report

Abstract

This project presents an ESP32-based Smart IoT Plant Irrigation System that automates irrigation using real-time sensor data. Soil moisture, rainfall, temperature, humidity, and water tank level are monitored continuously. The ESP32 processes the sensor readings and controls a water pump through a relay. The system supports both Automatic and Manual operating modes while providing real-time monitoring through the Blynk IoT platform.

1. Introduction

Traditional irrigation often wastes water because it relies on fixed schedules or manual observation. This project addresses that problem by automatically watering plants only when environmental conditions require it. The system improves water efficiency, reduces human effort, and protects the pump from operating without sufficient water.

2. Objectives

- Automate irrigation based on soil moisture.
- Prevent irrigation during rainfall.
- Monitor environmental conditions remotely.
- Protect the pump using water tank level detection.
- Allow remote and manual pump control through Blynk.

3. Hardware Components

ESP32 DevKit V1, Capacitive Soil Moisture Sensor, DHT22 Temperature and Humidity Sensor, FC-37 Rain Sensor, HC-SR04 Ultrasonic Sensor, 5V Relay Module, 5V Mini Water Pump, 16×2 LCD with I2C Backpack, Active Buzzer, Green LED, Red LED, Push Button, Breadboard, Jumper Wires and Power Supply.

4. Software Tools

Arduino IDE, ESP32 Board Package, Blynk IoT Platform, LiquidCrystal_I2C Library, DHT Library and Wi-Fi Library.

5. Working Principle

The ESP32 reads all sensors periodically. In Automatic Mode, the pump starts only when the soil moisture is below 35%, rainfall is not detected, and the water tank level is above 20%. In Manual Mode, the user can operate the pump from the Blynk mobile application. A 16×2 LCD displays operating mode, soil moisture, tank level, temperature, humidity and rain status using alternating screens.

6. Blynk Dashboard

Virtual Pins: V0 Soil Moisture, V1 Temperature, V2 Humidity, V3 Tank Level, V4 Rain Analog, V5 Pump Status, V6 Auto/Manual Mode, V7 Rain Status, V8 Manual Pump Control.

7. Testing

The system was tested by changing soil moisture, adding water to the rain sensor, changing the water level beneath the ultrasonic sensor, and using both manual and automatic pump control. The relay, LCD, LEDs, buzzer and Blynk dashboard responded correctly.

8. Results

The prototype successfully automated irrigation, reduced unnecessary watering, enabled remote monitoring, and protected the water pump when the tank level became too low.

9. Conclusion

The Smart IoT-Based Plant Irrigation System demonstrates an effective, low-cost irrigation solution using IoT technologies. The project integrates sensing, automation, remote monitoring, and safety mechanisms into a single platform suitable for small gardens and agricultural applications.

10. Future Improvements

Cloud database logging, weather forecast integration, AI-based irrigation scheduling, solar-powered operation, mobile notifications and historical analytics.

Pin Configuration

ESP32 Pin	Component
GPIO4	DHT22
GPIO34	Soil Moisture
GPIO32	Rain Analog
GPIO26	Rain Digital
GPIO5	HC-SR04 Trigger
GPIO18	HC-SR04 Echo
GPIO23	Relay
GPIO25	Green LED
GPIO33	Red LED
GPIO27	Buzzer
GPIO14	Push Button
GPIO21	LCD SDA
GPIO22	LCD SCL